ECPS 211 · Machine Learning & Data Mining · Winter 2026

Crypto Price Prediction

Rolling Backtesting · Baselines · LSTM · Ablation Study

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Dataset: ETH-USD (Ethereum) via Yahoo Finance · Project 11

OVERVIEW

Project Overview

01

Data

Yahoo Finance · yfinance
ETH-USD daily OHLCV
2017 – 2026, 3,038 rows

02

Baselines

Last Value (tomorrow = today)
7-Day Moving Average
MAE · RMSE · Dir. Acc.

03

Lag + Ridge

30 return lags + log_volume
+ volatility_14 · Pipeline
Optuna alpha (32 features)

04

LSTM

64→32 units · dropout
Early stopping · ReduceLROnPlateau
Next-day return target

05

Ablation

4 groups: lags, +volume,
+volatility, all 32
Permutation importance + PDP/ICE

06

Backtest

Ridge +120.6% (Sharpe 1.14)
LSTM −159.3% · Buy&Hold
ETH-USD bear market test

Dataset & Features

Source & Split

Provider: Yahoo Finance via yfinance

Asset: ETH-USD (Ethereum) — configurable

Range: 2017-01-01 to present

Rows: 3,038 daily records (OHLCV)

Preprocessing: Forward-fill, dropna

Timezone-naive datetime index

Split: Time-based only — no shuffle

Train 70% Val 15% Test 15%

(2,105 / 451 / 452 rows)

Derived Features

Returns r_t

$(P_t - P_{t-1}) / (P_{t-1} + \varepsilon)$

Day-over-day return (stationarity-friendly)

volatility_14

14-day rolling std(returns)

Backward-looking only

log_volume

log(1 + volume)

Log-transformed volume

Return lags 1–30

$r_{t-1}, r_{t-2}, \ldots r_{t-30}$

30 return lags + 2 extra features (32 total)

METHODS

Evaluation Framework & Metrics

MAE

Mean Absolute Error

mean $|y_i - \hat{y}_i|$

Magnitude of price error

RMSE

Root Mean Squared Error

$\sqrt{mean(y_i - \hat{y}_i)^2}$

Penalizes large errors more

Dir. Acc.

Directional Accuracy

% correct up/down sign

Practical trading signal

Methodology Principles

- Time-based splits only — no data shuffling to prevent future leakage
- Expanding-window walk-forward: retrain Ridge on all past data per test window
- Optuna (Bayesian) for Ridge alpha; objective: validation MAE on returns
- Same test period for all models: baselines, Ridge, and LSTM
- Ablation: 4 feature groups (lags only, +volume, +volatility, all 32)

Baseline Models

Baseline 1 — Last Value

$$\hat{P}_{t+1} = P_t$$

Tomorrow's price equals today's price.
No information from volume or direction.
Dir. Acc. is not meaningful (predicts no change → often 0%).

$82.21	$117.91
MAE (ETH-USD)	RMSE (ETH-USD)

✓ Best on MAE & RMSE across all assets
✗ Directional accuracy is not meaningful

Baseline 2 — 7-Day Moving Average

$$\hat{P}_{t+1} = (1/7)\,\Sigma_{i=0}^{6}\,P_{t-i}$$

Smoothed price over 7 prior days.
Captures local trend, filters single-day noise.
Dir. Acc. is meaningful (predicts a direction).

$138.74	51.58%
MAE	Dir. Acc.

✓ Best directional accuracy among baselines (51.58%)
✗ Higher MAE/RMSE than last value (more smoothing error)

LAG-FEATURE MODEL

Ridge Regression with Lag Features

Feature Engineering & Pipeline

Features (32 total): 30 return lags + log_volume + volatility_14

Scaler: StandardScaler on all 32 columns via ColumnTransformer

Model: Ridge regression (linear; regularized)

Hyperparameter search: Optuna (Bayesian) over Ridge alpha; objective: validation MAE on returns
Best alpha applied to test; permutation importance and PDP/ICE computed on test set

Target: Next-day return (stationarity-friendly); price via $\hat{P}_{t+1} = P_t(1 + \hat{r}_t)$

Test Set Results (ETH-USD)

$83.72
MAE

$118.41
RMSE

54.16%
Directional Accuracy

Rolling Backtest Results

Train on past only → predict next day → roll forward
Retrain on all past data before each test window (expanding window)

Mean ± std of MAE and Dir.Acc across multiple test windows

Expanding window is stricter; Dir.Acc 55.55% ± 5.49%

LSTM Architecture & Results

Architecture

Target: Next-day return r_t (not raw price)
Converted: $\hat{P}_{t+1} = P_t \cdot (1 + \hat{r}_t)$

Sequence: Length 30; 3 features per timestep (ret, log_volume, volatility_14)

Scaling: StandardScaler fit on train; applied to val/test

Layers: LSTM(64) → LSTM(32) → Dense(1) + Dropout(0.2)

Training: MSE loss · batch 32 · early stopping on val loss · ReduceLROnPlateau

Test Set Results (ETH-USD)

$82.13
MAE

$117.89
RMSE

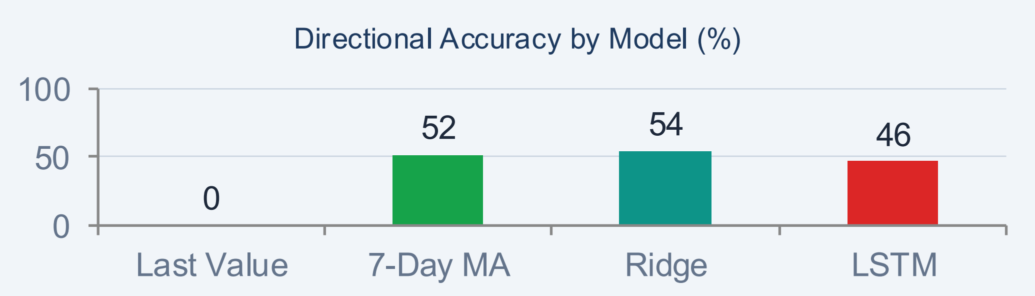
46.32%
Directional Accuracy

Key observation: LSTM MAE ($82.13) is close to Last Value; Dir.Acc (46.32%) is below 50% — no consistent edge over simpler models on ETH-USD.

RESULTS

Full Model Comparison (ETH-USD, Test Set)

Model	MAE	RMSE	Dir. Acc.	Strength
Last Value	**$82.21 ★**	**$117.91 ★**	0% *	Best magnitude
7-Day MA	$138.74	$192.77	**51.58%**	Best baseline dir.
Ridge	$83.72	$118.41	54.16% ★	Best direction
LSTM	$82.13	$117.89	46.32%	Below 50% dir.



* *Last value predicts no change; directional accuracy = 0% is not meaningful.*

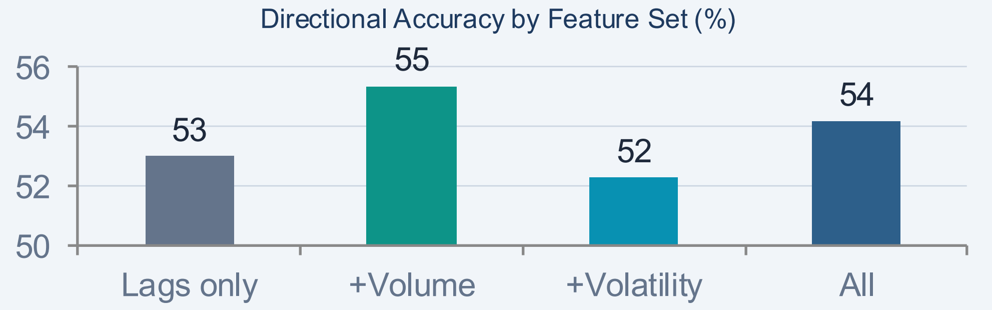
Key Takeaways

- Last value wins MAE/RMSE — hard to beat
- Ridge wins direction at 54.16%
- 7-Day MA best baseline dir. (51.58%); LSTM below 50% (46.32%)

ABLATION STUDY

Feature Ablation — What Actually Helps?

Feature Set	# Features	MAE	RMSE	Dir. Acc.
Lags only	30	**0.028193 ★**	—	52.97%
Lags + Volume	31	0.028550	—	**55.34% ★**
Lags + Volatility	31	0.028210	—	52.26%
All (Lags + Vol + Volat.)	32	0.028497	—	54.16%



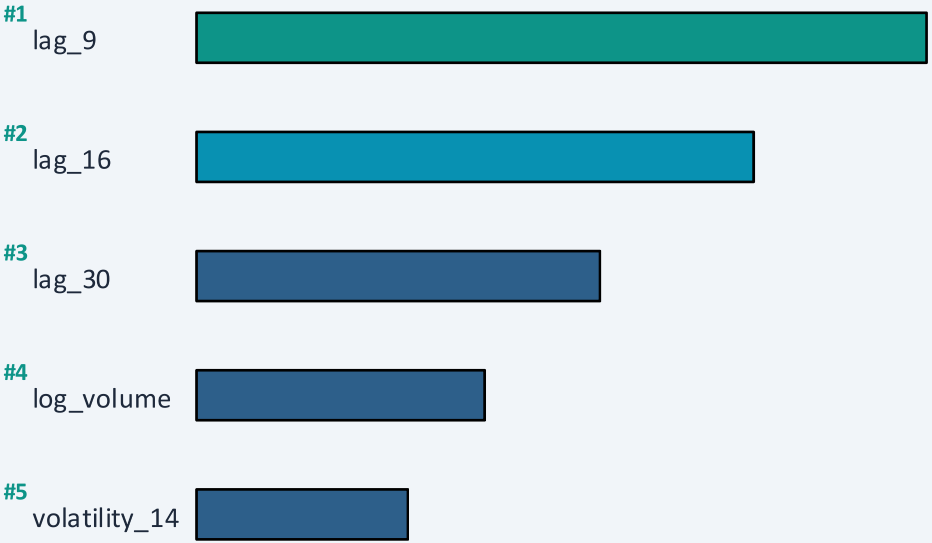
Ablation Insight

- Lags only best MAE (0.028193)
- Adding volume: largest direction gain (+1.9 pp vs lags only)
- +Volatility alone adds little over lags
- +Volume gives the largest direction gain at small cost to magnitude

Permutation Importance · PDP / ICE

Permutation Importance (Best Lag+Ridge)

Shuffle each feature 10× · measure MAE increase · ETH-USD test set



Top-3 features are lag_9, lag_16, and lag_30 (mid-range lags dominate). Volume and volatility rank 4th/5th — price momentum lags matter most for ETH-USD.

PDP / ICE Insights

lag_9

Roughly linear PDP — higher lag_9 → higher prediction. Effect is homogeneous across ICE curves (Ridge is linear in features).

log_volume

Flatter PDP — volume has a weaker marginal effect on price than lag features, but matters for directional accuracy (ablation).

ICE curves for lag_9 show similar slope across samples — the linear effect is consistent, not sample-specific.

ERROR ANALYSIS

Where Does the Model Fail?

Long-Tailed Residuals

Residual histogram is centered near zero but has heavy tails — large errors occur in both directions, indicating the model cannot handle sharp price moves.

Clustered Errors Over Time

Errors are not uniformly distributed across time. They cluster in volatility regimes — calm periods have smaller errors; turbulent periods have large errors.

High-Volatility Failure

|Residual| vs volatility_14 plot shows a positive correlation — larger absolute errors occur when the 14-day rolling volatility is high. Model performs worst in volatile markets.

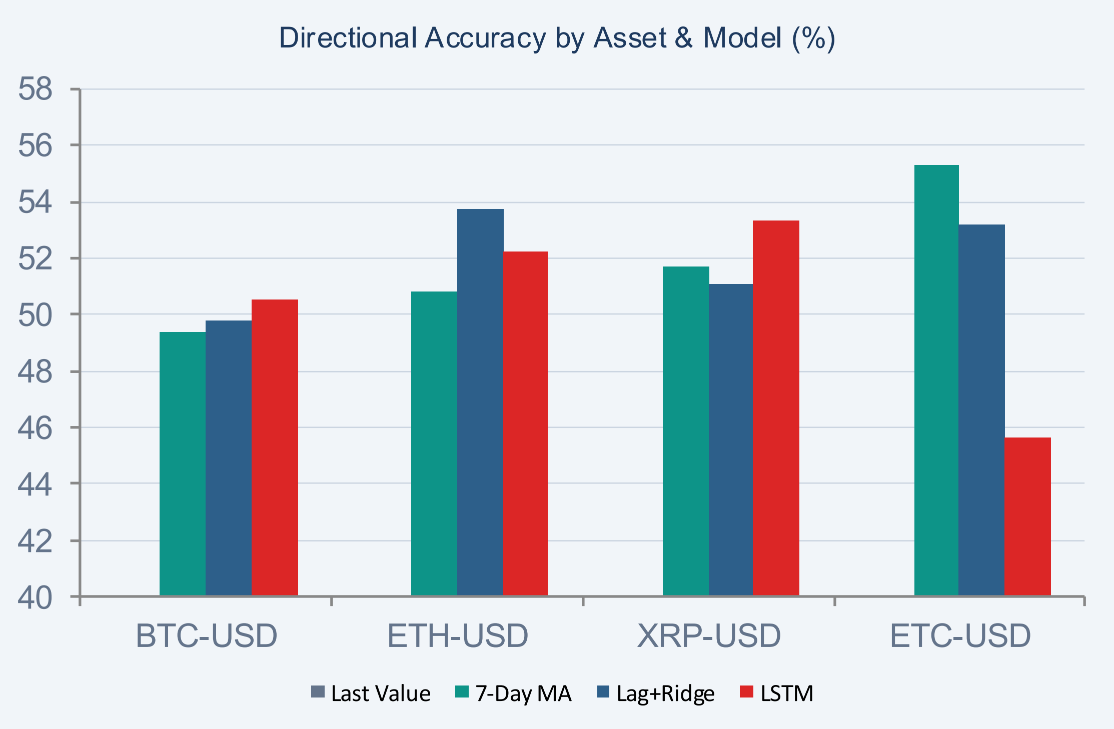
Smoothed Extrapolation

Lag+Ridge behaves like a smoothed extrapolation of recent price lags. It systematically under- or over-predicts during sharp moves and mean-reverting spikes.

Conclusion: Lag+Ridge (and simpler models) are smoothing extrapolators — they fail most in high-volatility regimes and on large price moves.

CROSS-ASSET

Generalization Across Cryptocurrencies



BTC-USD

Best dir.: LSTM dir. edge (50.5%)

All models near 50%; last value best MAE

ETH-USD

Best dir.: Ridge (54.16%)

Slightly more directionally predictable

XRP-USD

Best dir.: LSTM (53.3%)

Only asset where LSTM wins direction

ETC-USD

Best dir.: 7-Day MA (55.3%)

Best directional result; LSTM < 50%

Last value wins MAE/RMSE on every asset. ETH-USD: Ridge leads direction (54.16%); 7-Day MA best among baselines (51.58%).

EXPANDING-WINDOW EVALUATION

Expanding-Window Walk-Forward — Ridge

How Expanding-Window Evaluation Works

Step 1	Train: Days 1–531	→ Day 532
Step 2	Train: Days 1–532	→ Day 533
Step 3	Train: Days 1–533	→ Day 534
…	Train: …	→ …

Minimum training window required; retrain Ridge before each test window

Fixed Split vs. Expanding-Window — Ridge (ETH-USD)

Evaluation	Fixed-Split Test	Expanding-Window
MAE	$83.72	0.030806
RMSE	$118.41	—
Dir. Acc.	54.16%	55.55% ± 5.49%

CONCLUSION

Key Findings

Baselines are competitive: Last value achieves best MAE/RMSE across all assets — a surprisingly strong benchmark.

Ridge on direction: Best ML model — 54.16% Dir.Acc on ETH-USD test set (best of all models).

LSTM: LSTM Dir.Acc 46.32%; competitive MAE ($82.13) but below 50% direction.

Ablation (4 groups): Volume improves direction most (+2.37 pp, from 52.97% to 55.34%). 32 features total — lags + log_volume + volatility_14.

Error analysis: All models fail in high-volatility regimes and on large moves. Expanding-window is stricter than fixed-split.

Reproducibility: notebooks/Crypto_Colab_AllInOne.ipynb · Set ASSET (e.g. ETH-USD) and run all cells